

Statistical Programming Lab- ADL47

Part A

1. Suppose that `queue <- c ("Steve", "Russell", "Alison", "Liam")` and that `queue` represents a supermarket queue with Steve first in line. Using R expressions update the supermarket queue as successively:
 - a. Barry arrives
 - b. Steve is served
 - c. Pam takes her way to the front with one item
 - d. Barry gets impatient and leaves
 - e. Alison gets impatient and leaves
 - f. Using the function `which(x)`, find the position of Russell in the queue
2. Plot the graph of linear equation, Non-linear equation: square, cubic, square root, etc.,
3. The performance of a student in 3rd semester CSE is as given as {SUB1, SUB2, SUB3, SUB4, SUB5, SUB6} with score {91, 73, 65, 45, 54, 32}. Draw the Bar chart and pie chart to indicate the performance of the student. Find the average marks and discuss qualitatively about the performance in each subject.

Part B

1. You are working as a cashier at a grocery store. Your task is to create a program that simulates the checkout process for a customer's shopping cart. The program should calculate the total cost of the items, including tax, and provide a detailed receipt.
 - i. Define a list of products, each represented as a dictionary with keys: "name", "price".
 - ii. Allow the cashier to input the products in the customer's shopping cart, including the name, price, and quantity of each item.
 - iii. Calculate the subtotal (price * quantity) for each item and display a detailed receipt with product names, quantities, prices, and subtotals.
 - iv. Calculate the total cost of the items in the cart before tax.
 - v. Apply a tax rate (e.g., 8%) to the total cost to calculate the tax amount.
 - vi. Calculate the final total cost by adding the tax amount to the total cost before tax.